NTFS

NEW TECHNOLOGY FILE SYSTEM

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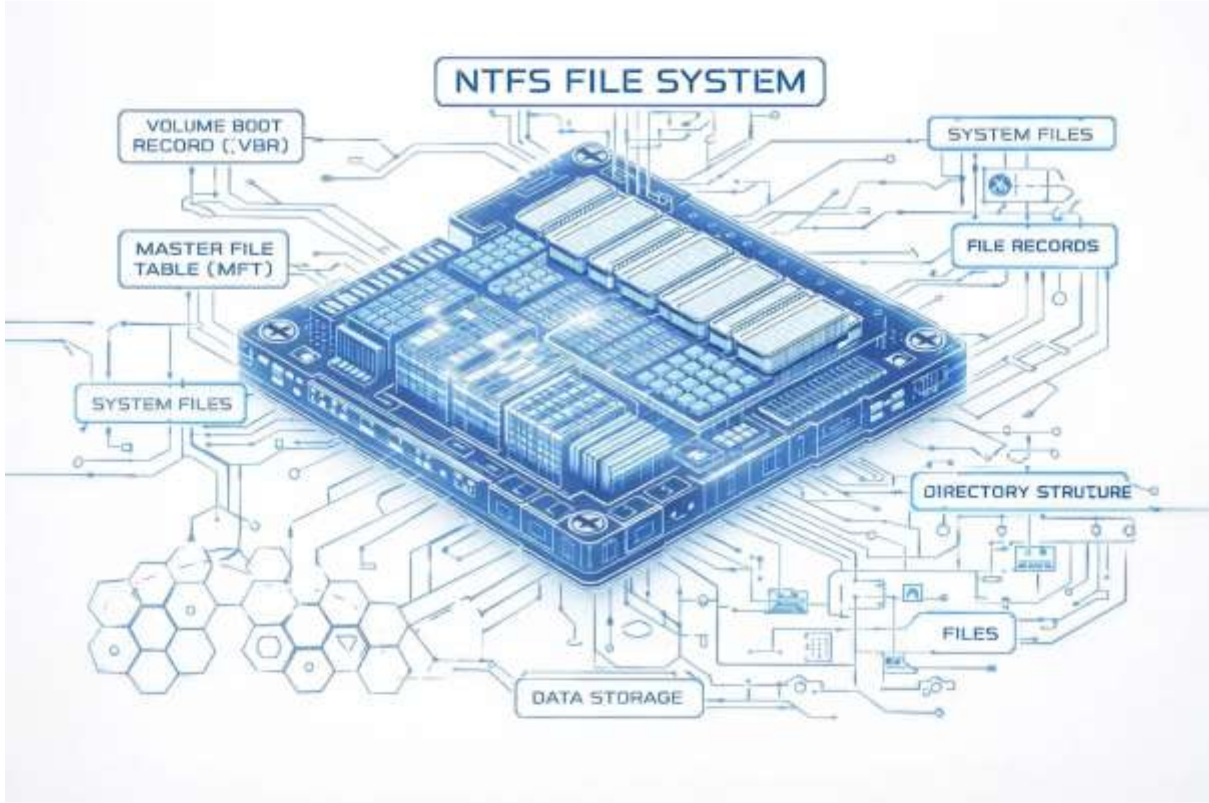
- **NTFS (New Technology File System)** is the **file system used by Windows** to organize, store, secure, and manage files and folders on storage devices like HDDs and SSDs


NTFS FILE SYSTEM
VOLUME BOOT RECORD (VBR)
SYSTEM FILES
MASTER FILE TABLE (MFT)
FILE RECORDS
SYSTEM FILES
DIRECTORY STRUTURE
FILES
DATA STORAGE

WHAT IS A FILE SYSTEM?

- A **File System** is a method used by an operating system to **store, organize, manage, and retrieve files** on a storage device such as a hard disk, SSD, or USB drive.

WHY IS NTFS IMPORTANT IN CYBER FORENSICS?

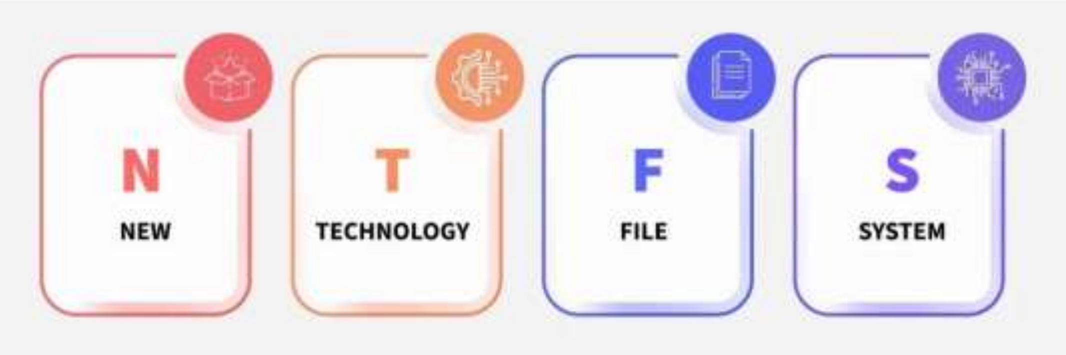
- Because criminals also use computers.
- Every document, photo, email, or video they create is stored using the file system.
- So forensic investigators study NTFS to understand:

📂 Where files are stored

□ How deleted files can be recovered

🔒 How encrypted files are protected

🎁 How compressed files are stored



FEATURES OF NTFS

- NTFS is more than just a place to store files. It provides several powerful features
    1. **NTFS Compression**
    2. **NTFS Encryption (EFS)**
    3. **Deleting NTFS Files**

NTFS COMPRESSION

- **NTFS Compression** is a feature that **reduces the size of files and folders so they use less storage space while keeping the original data unchanged.**

When a file is saved:

1. Windows compresses the file.
2. The compressed file uses less storage space.
3. When you open the file, Windows automatically decompresses it.
4. When you save it again, Windows compresses it automatically.

The user doesn't notice this process because it happens in the background.

NTFS COMPRESSION

- In Cyber Forensics

Investigators must examine compressed files because they may contain important evidence such as documents, emails, or images.

Example :

- A project folder is **100 MB**.
- After NTFS Compression,
- it may occupy only **60 MB**.
- The contents remain exactly the same.

NTFS ENCRYPTION (EFS)

- **NTFS Encryption**, also known as the **Encrypting File System (EFS)**, is a security feature of NTFS that **encrypts files and folders so that only authorized users with the correct encryption key can access them.**

➢ When a file is encrypted:

1. Windows converts the file into an unreadable format.
2. An encryption key is created.
3. Only the user with the correct key can open and read the file.

➢ Anyone else will only see unreadable data.

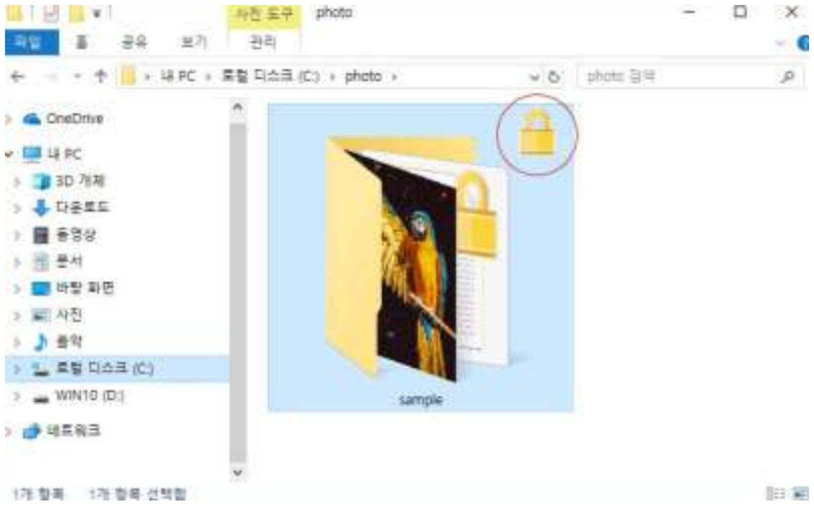


사진 도구
photo
파일
홈
공유
보기
관리
내 PC › 로컬 디스크 (C:) › photo ›
photo 검색
OneDrive
내 PC
3D 개체
다운로드
동영상
문서
바탕 화면
사진
음악
로컬 디스크 (C:)
WIN10 (D:)
네트워크
sample
1개 항목
1개 항목 선택함

NTFS ENCRYPTION

In Cyber Forensics,

- Encrypted files are difficult to examine.
- Investigators often need the user's encryption key or authorized access before they can read the evidence.

Example

- Without encryption:
- 📄 Salary_Report.docx
- Anyone using the computer can open it.
- After encryption:
- 🔒 Salary_Report.docx

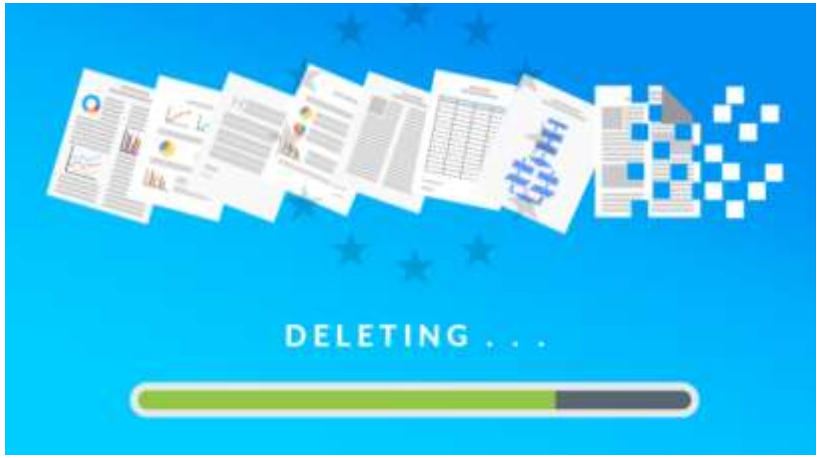
Only the authorized user can read it.

DELETING NTFS FILES

- When a file is deleted in NTFS, the file is not immediately erased from the storage device. Instead, NTFS marks the space occupied by the file as available for reuse, while the actual data remains until it is overwritten by new data.

When you delete a file:

1. Windows removes the file's entry from the file system.
2. The storage space is marked as free.
3. The actual data remains on the disk.
4. The data disappears only when new information overwrites it.



NTFS DELETION

In Cyber Forensics

- Recovering deleted files is one of the most important tasks in digital forensics because deleted documents, emails, chats, and photos can provide valuable evidence in criminal investigations.

Example:

- You delete a photo called **Vacation.jpg**.
- Although you cannot see it anymore,
- the photo may still exist on the hard disk.
- A forensic recovery tool can often recover it if it hasn't been overwritten.

THANK YOU